INCH-POUND

MIL-M-28787/256
AMENDMENT 2
28 April 1993
SUPERSEDING
AMENDMENT 1
6 July 1983

MILITARY SPECIFICATION

MODULES, STANDARD ELECTRONIC

FLIP-FLOP, D-TYPE, DIGITAL, KEY CODE BMD

This amendment forms a part of Military Specification MIL-M-28787/256(EC), dated 24 July 1981, and is approved for use by all Departments and Agencies of the Department of Defense.

PAGES 4, 5, 6, 7, 8, AND 9

*TABLE I: Delete and substitute new table I as printed on pages 4, 5, 6, 7, 8, 9, 10, 11, and 12 of this amendment.

PAGE 10

*4.4.2: Delete and substitute the following:

"4.4.2 Order of testing. Test conditions A through SS shall be performed in any order with the following exceptions. Test conditions A, B, and C are stress tests and shall be performed only at 25 ± 5°C. If test conditions A, B, or C are part of a sequence of tests, the order of testing shall be as follows: A, B, and C where applicable, followed by the remaining tests in the sequence."

*4.5.1, fifth and sixth lines: Delete "B through J," and substitute "D through J,".

4.5.1, eighth line: Delete "100°C" and substitute "85°C".

*4.5.3, third line: Delete "B through J," and substitute "D through J,".

*4.5.3, sixth line: Delete "A through SS" and substitute "D through SS".

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PAGE 11

6.4, Output drive current (V_{DD} = 5 V): Delete "100°C" (two places) and substitute "85°C" (two places).

*6.4, Input current low level: Delete "V_{DD} = 18 V)" and substitute "V_{DD} = 15 V)".

*6.4, Input current high level: Delete "(V_{IH} = 18 V, V_{DD} = 18 V)" and substitute "(V_{IH} = 15 V, V_{DD} = 15 V)".

PAGE 13

*FIGURE 2, ITEM P07 box: Delete "18 V ± 1%" and substitute "15 V ± 1%".

PAGE 14

*FIGURE 3: Delete and substitute new figure 3 as printed on page 13 of this amendment.

PAGE 16

*FIGURE 5, ITEM P07 box: Delete "18 V ± 10%" and substitute "15 V ± 10%".

PAGE 23

FIGURE 7: Delete and substitute new figure 7 as printed on page 14 of this amendment.

PAGE 26

FIGURE 8: Delete and substitute new figure 8 as printed on page 15 of this amendment.

PAGE 29

FIGURE 9: Delete and substitute new figure 9 as printed on page 16 of this amendment.

PAGE 32

FIGURE 10: Delete and substitute new figure 10 as printed on page 17 of this amendment.

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PAGE 35

*FIGURE 11, ITEM P07 box: Delete "18 V ± 1%" and substitute "15 V ± 1%".

PAGE 38

*FIGURE 12, ITEM P07 box: Delete "18 V ± 1%" and substitute "15 V ± 1%".

PAGES 82 AND 83

TABLE XXXVI, Table heading, Connect columns: Add "A to pin" (two places) in the left column and "B to pin" (two places) in the right column.

NOTE: The margins of this amendment are marked with an asterisk or vertical lines to indicate where changes (additions, modifications, corrections, deletions) from the previous amendment were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the last previous amendment.

Custodians:
Army - ER
Navy - SH
Air Force - 85

Review activities:
Army - AT
Navy - AS, MC
Air Force - 13, 19
DLA - ES

Preparing activity:
Navy - NW

(Project 5963-0208)

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TABLE I. Module electrical requirements.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
A	Isolation													
	Initial	III	S	10	∞	10	∞	10	∞	10	∞	10	∞	MΩ
	End-of-life	III	S	1	∞	1	∞	1	∞	1	∞	1	∞	MΩ
B1-B24	Input clamp diode voltage, negative	IV	S	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	-0.4	-6.0	V
C1-C24	Input clamp diode voltage, positive	V	S	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	0.4	1.5	V
D1-D12	Static output voltage, low level, IOL = 0	VI	X	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	0.00	0.05	V
E1-E12	Static output voltage, high level, IOH = 0	VII	X	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	4.95	5.05	V
F1-F12	Static output voltage, low level, VDD = 5 V	VIII	X	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	0.0	0.4	V
G1-G12	Static output voltage, high level, VDD = 5 V	IX	X	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	4.6	5.0	V
H1-H12	Static output voltage, low level, VDD = 15 V	X	X	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	0.0	1.5	V
J1-J12	Static output voltage, high level, VDD = 15 V	XI	X	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	13.5	15.0	V

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
K1-K24	Input current, low level	XII	X	0	-200	---	---	0	-200	---	---	0	-200	nA
L1-L24	Input current, high level	XIII	X	0	200	---	---	0	200	---	---	0	200	nA
M1	Static power supply current, low level	XIV	X	0	60	0	60	0	590	0	60	0	971	μA
M2	Static power supply current, high level	XIV	X	0	60	0	60	0	590	0	60	0	971	μA
N1-N12	Transition time (tTLH), CL = 50 pF, VDD = 5 V	XV	S	36	360	36	360	40	400	27	275	43	425	ns
N13-N24	Transition time (tTLH), CL = 150 pF, VDD = 5 V	XV	Q	84	840	84	840	93	930	64	640	99	993	ns
P1-P12	Transition time (tTHL), CL = 50 pF, VDD = 5 V	XVI	S	20	200	20	200	22	220	15	150	23	236	ns
P13-P24	Transition time (tTHL), CL = 150 pF, VDD = 5 V	XVI	Q	68	680	68	680	75	750	51	515	80	803	ns
Q1-Q12	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XVII	S	35	350	35	350	38	385	26	265	41	413	ns

-See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test conditions	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
Q13-Q24	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XVII	Q	55	550	55	550	61	610	42	420	65	651	ns
R1-R12	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XVIII	S	35	350	35	350	38	385	26	265	41	413	ns
R13-R24	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XVIII	Q	55	550	55	550	61	610	42	420	65	651	ns
S1-S6	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XIX	S	35	350	35	350	38	385	26	265	41	413	ns
S7-S12	Propagation delay time (tPLH), CL = 50 pF, VDD = 5 V	XIX	S	45	450	45	450	50	500	34	340	53	531	ns
S13-S18	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XIX	Q	55	550	55	550	61	610	42	420	65	651	ns
S19-S24	Propagation delay time (tPLH), CL = 150 pF, VDD = 5 V	XIX	Q	65	650	65	650	72	720	49	495	76	767	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
T1-T6	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XX	S	35	350	35	350	38	385	26	265	41	413	ns
T7-T12	Propagation delay time (tPHL), CL = 50 pF, VDD = 5 V	XX	S	45	450	45	450	50	500	34	340	53	531	ns
T13-T18	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XX	Q	55	550	55	550	61	610	42	420	65	651	ns
T19-T24	Propagation delay time (tPHL), CL = 150 pF, VDD = 5 V	XX	Q	65	650	65	650	72	720	49	495	76	767	ns
U1-U12	Transition time (tTLH), CL = 50 pF, VDD = 10 V	XXI	Q	18	180	18	180	20	200	13	135	21	213	ns
V1-V12	Transition time (tTHL), CL = 50 pF, VDD = 10 V	XXII	Q	10	100	10	100	11	110	7	75	12	119	ns
W1-W12	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXIII	Q	15	150	15	150	16	165	11	115	17	178	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
X1-X12	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXIV	Q	15	150	15	150	16	165	11	115	17	178	ns
Y1-Y6	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXV	Q	15	150	15	150	16	165	11	115	17	178	ns
Y7-Y12	Propagation delay time (tPHL), CL = 50 pF, VDD = 10 V	XXV	Q	20	200	20	200	22	220	15	150	23	236	ns
Z1-Z6	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXVI	Q	15	150	15	150	16	165	11	115	17	178	ns
Z7-Z12	Propagation delay time (tPLH), CL = 50 pF, VDD = 10 V	XXVI	Q	20	200	20	200	22	220	15	150	23	236	ns
AA1-AA12	Transition time (tTLH), CL = 50 pF, VDD = 15 V	XXVII	Q	13	130	13	130	14	145	10	100	15	154	ns
BB1-BB12	Transition time (tTHL), CL = 50 pF, VDD = 15 V	XXVIII	Q	8	80	8	80	9	90	6	60	10	96	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Test end	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
							0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
					Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
CC1	CC12	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXIX	Q	10	100	10	100	11	110	7	75	12	119	ns
DD1	DD12	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXX	Q	10	100	10	100	11	110	7	75	12	119	ns
EE1	EE6	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXXI	Q	10	100	10	100	11	110	8	80	12	116	ns
EE7	EE12	Propagation delay time (tPLH), CL = 50 pF, VDD = 15 V	XXXI	Q	15	150	15	150	16	165	11	115	17	178	ns
FF1	FF6	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXXII	Q	10	100	10	100	11	110	8	80	12	119	ns
FF7	FF12	Propagation delay time (tPHL), CL = 50 pF, VDD = 15 V	XXXII	Q	15	150	15	150	16	165	11	115	17	178	ns
GG1	GG12	Maximum shift frequency check	XXXIII	S	900	1100	900	1100	900	1100	900	1100	900	1100	ns

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
HH	Power supply filter capacitance	XXXIV	S	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	0.007	0.013	µF
JJ1-JJ2	C2 and C3 capacitance	XXXV	X	176	264	176	264	176	264	176	264	176	264	pF
KK1-KK24	Input capacitance	XXXVI	Q	2	15	2	15	2	15	2	15	2	15	pF
LL1-LL3	Functional test	XXXVII	X	<— See table XXXVII —>										
MM1-MM12	Data pulse set time (tSLH), VDD = 5 V, CL = 50 pF	XXXVIII	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
MM13-MM24	Data pulse set time (tSHL), VDD = 5 V, CL = 50 pF	XXXVIII	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
NN1-NN12	Data pulse hold time (tHHL), VDD = 5 V, CL = 50 pF	XXXIX	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
NN13-NN24	Data pulse hold time (tHLH), VDD = 5 V, CL = 50 pF	XXXIX	S	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS

- See footnotes at end of table.

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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
PP11	PP12 Data pulse set time (tSLH), VDD = 10 V, CL = 50 pF	XL	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
PP11	-PP24 Data pulse set time (tSHL), VDD = 10 V, CL = 50 pF	XL	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
QQ11	QQ12 Data pulse hold time (tHHL), VDD = 10 V, CL = 50 pF	XLI	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
QQ11	-QQ24 Data pulse hold time (tHLH), VDD = 10 V, CL = 50 pF	XLI	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
RR11	-RR12 Data pulse set time (tSLH), VDD = 15 V, CL = 50 pF	XLII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
RR11	-RR24 Data pulse set time (tSHL), VDD = 15 V, CL = 50 pF	XLII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS
SS11	-SS12 Data pulse hold time (tHLH), VDD = 15 V, CL = 50 pF	XLIII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS

- See footnotes at end of table.

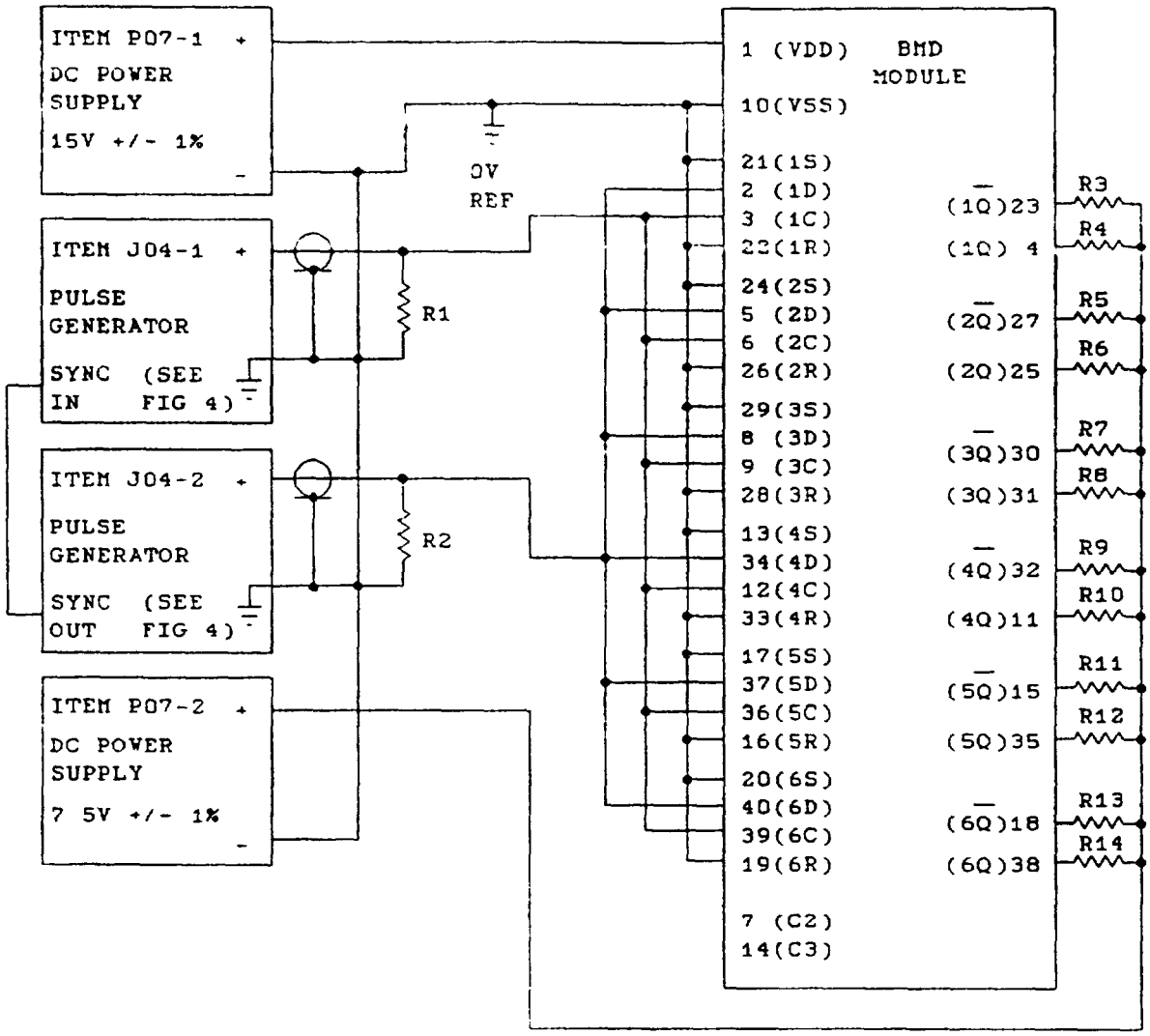
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TABLE I. Module electrical requirements - Continued.

Test cond	Parameter	Ref test table	Test type 1/-3/	25 ± 5°C 4/		Class I 4/				Class II 4/				Unit
						0 ± 5°C		60 ± 5°C		-55 ± 5°C		85 ± 5°C		
				Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	
SS13 SS24	Data pulse hold time (tHHL), VDD = 15 V, CL = 50 pF	XLIII	Q	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	3.6	4.4	µS

1/ "Q" parameters shall be 100 percent tested during module acceptance tests.
2/ "S" parameters shall be sample acceptance tested (1 percent or two modules from each inspection lot, whichever is greater) at 25°C.
3/ "Q" parameters shall be tested during qualification testing only.
4/ Unless otherwise specified, limits are for initial and end-of-life.

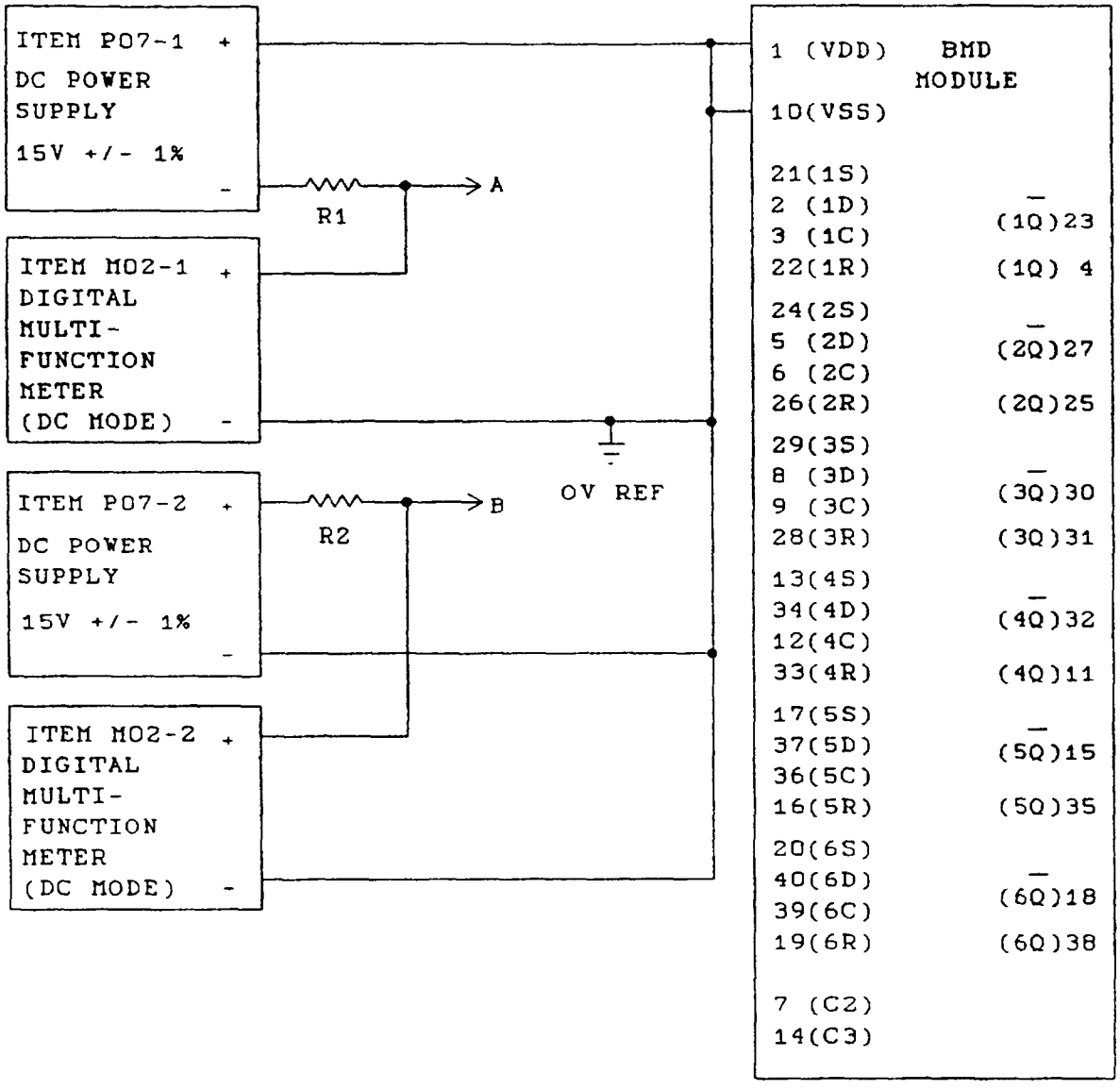
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NOTE: R1 = R2 = 51.1 Ω ± 5%, 5 W.
R3-R14 = 2.55 kΩ ± 5%, 1/4 W.

FIGURE 3. Component temperature test circuit.

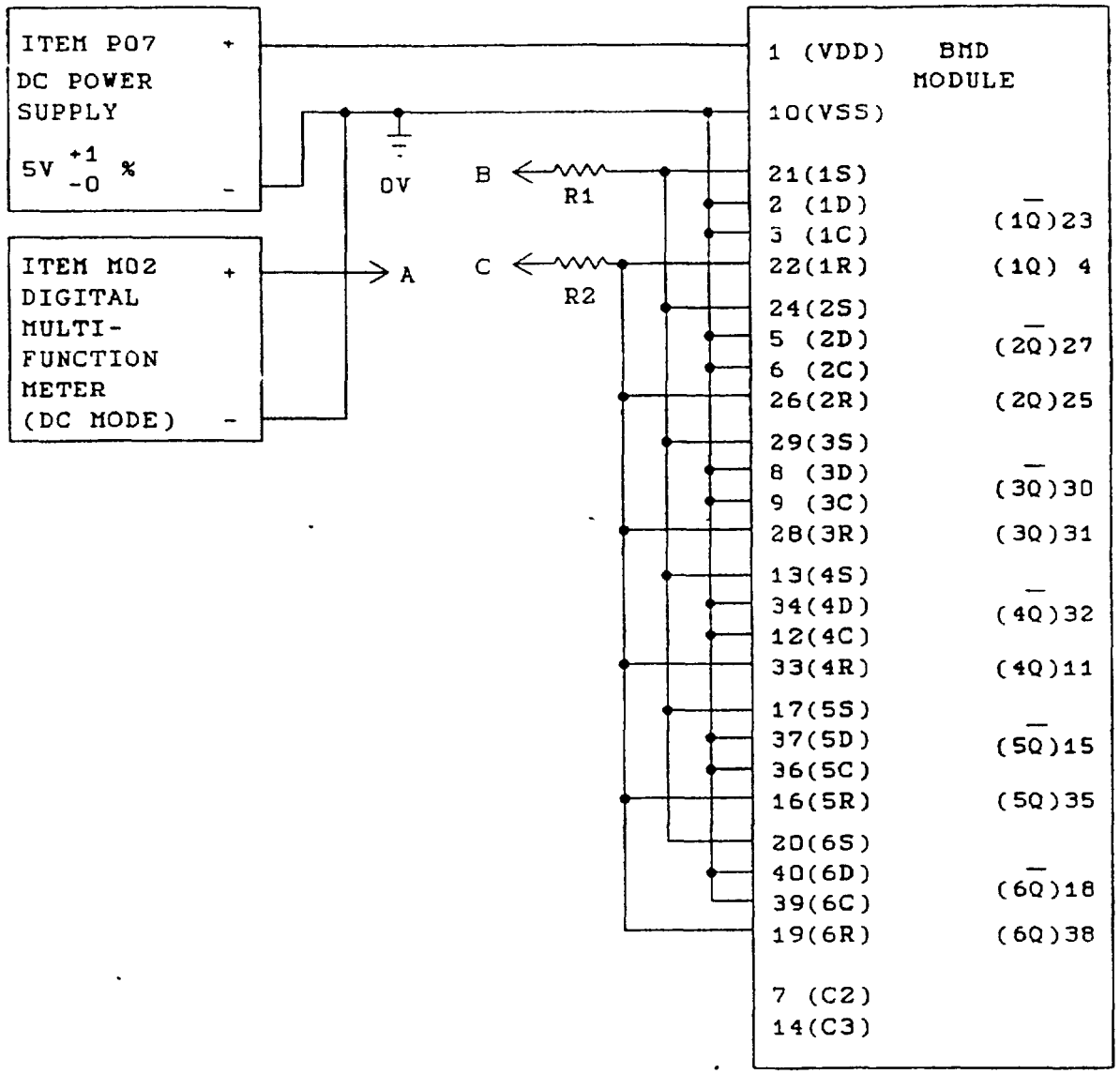
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NOTES:
1. R1 = 9.09 kΩ ± 1%, 1/8 W.
2. R2 = 13.7 kΩ ± 1%, 1/8 W.
3. Since items P07-1 and P07-2 and items M02-1 and M02-2 are not simultaneous, only one item P07 and one item M02 is required.

FIGURE 7. Input clamp diode voltage (negative and positive) test circuit for tables IV and V.

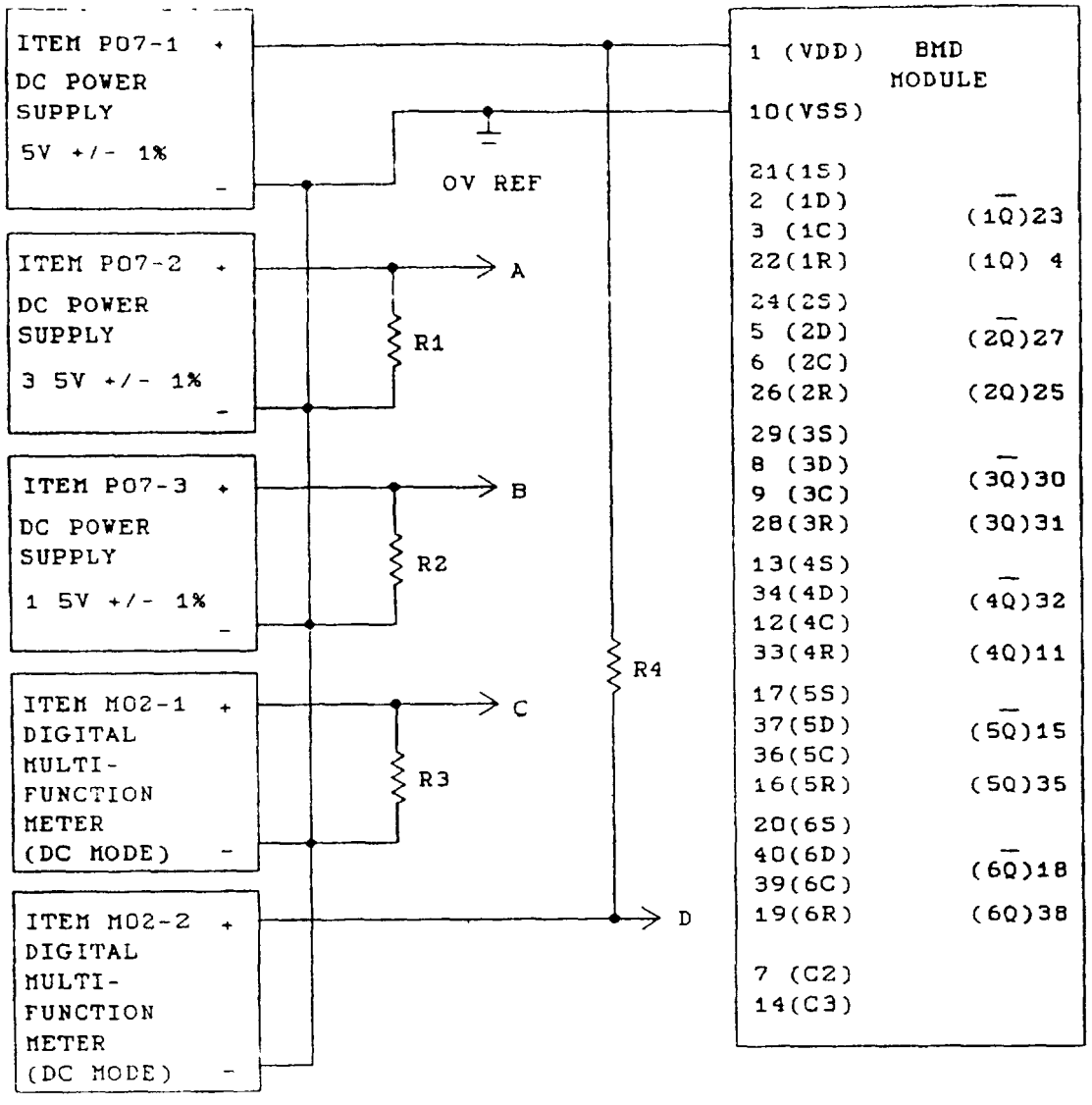
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NOTE: R1 = R2 = 1.5 kΩ ± 1%, 1/4 W.

FIGURE 8. Static output voltage (low-level, IOL = 0 and high-level, IOH = 0) test circuit for tables VI and VII.

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NOTES:
1. R1 = R2 = 51.1 Ω ± 5%, 1/8 W.
2. R3 = @ 0°C and 25°C 23.2 kΩ ± 1%, 1/8 W.
   = @ 60°C 27.4 kΩ ± 1%, 1/8 W.
   = @ -55°C 18.2 kΩ ± 1%, 1/8 W.
   = @ 85°C 32.4 kΩ ± 1%, 1/8 W.
3. R4 = @ 0°C and 25°C 9.09 kΩ ± 1%, 1/8 W.
   = @ 60°C 10.5 kΩ ± 1%, 1/8 W.
   = @ -55°C 7.15 kΩ ± 1%, 1/8 W.
   = @ 85°C 12.7 kΩ ± 1%, 1/8 W.

FIGURE 9. Static output voltage (low-level, VDD = 5 V and high-level, VDD = 5 V) test circuit for tables VIII and IX.

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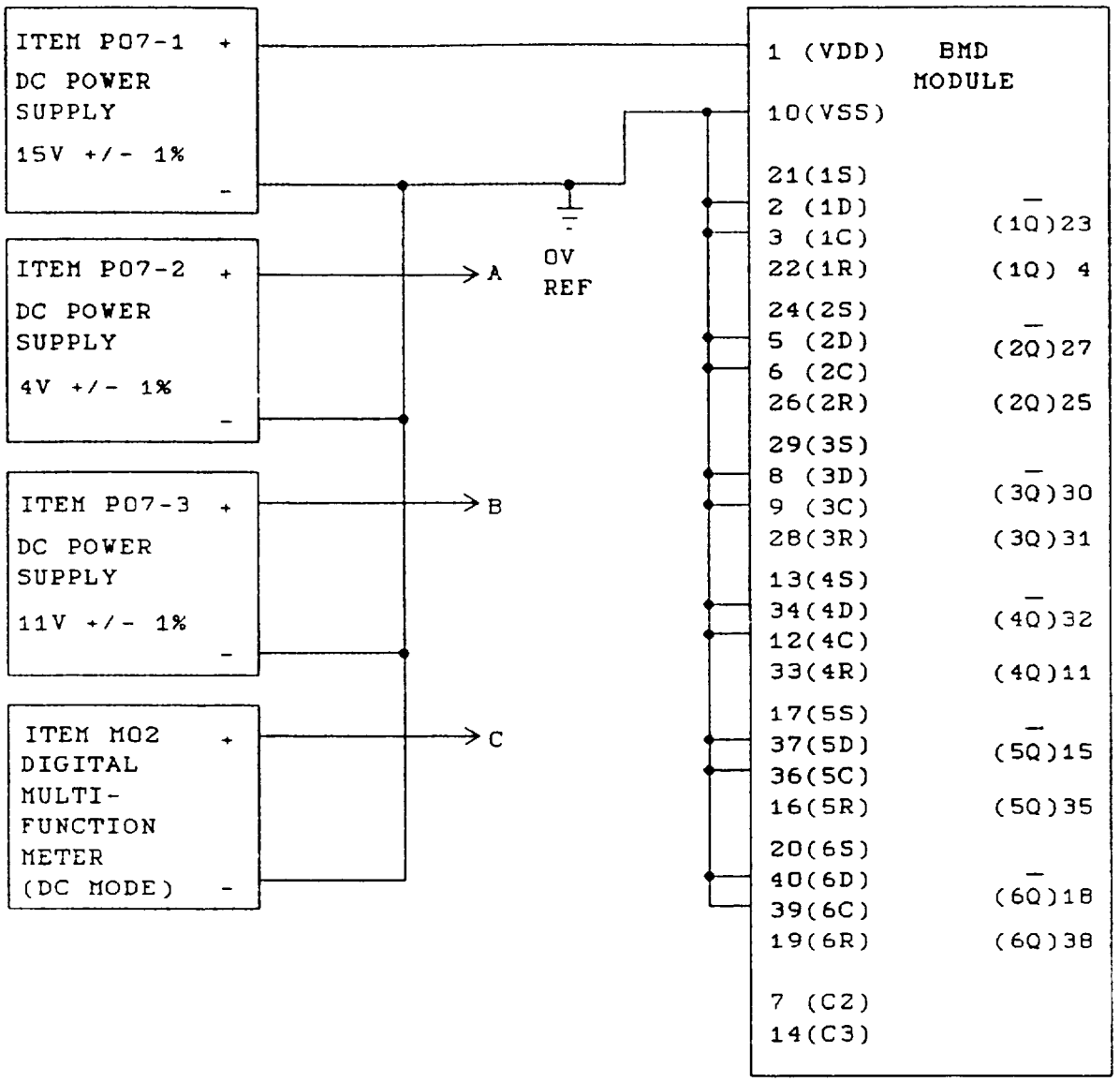


FIGURE 10. Static output voltage (low-level, VDD = 15 V and high-level, VDD = 15 V) test circuit for tables X and XI.

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